

What Spatial Fine-Tuning Actually Changes: A Multi-Seed Decomposition of Accuracy, Evidence Dependence, and Semantic Consistency in Vision–Language Models

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Abstract

001 Benchmark accuracy is the standard currency for re-
002 porting spatial fine-tuning of vision–language models
003 (VLMs), yet an accuracy gain alone does not say what
004 an adaptation actually changed: the same update may
005 alter how strongly the model relies on the correct im-
006 age, how coherently it applies relational semantics, and
007 how it responds to a causal image transformation—
008 independent components moving in different directions
009 at different rates. We introduce a preregistered eval-
010 uation protocol that measures spatial fine-tuning as
011 a jointly reported vector of capability changes rather
012 than a scalar score: benchmark gain ΔA , correct-image
013 dependence ΔG (the normal-minus-shuffled-image ac-
014 curacy gap relative to zero-shot), semantic pair con-
015 sistency ΔC (complement and facing-antonym flip-law
016 compliance), and Tier-C behavior under global hor-
017 izontal reflection (transformed accuracy, pair consis-
018 tency, joint correctness). We apply the protocol to or-
019 dinary LoRA fine-tuning on Visual Spatial Reasoning
020 as a controlled intervention, across two backbone fam-
021 ilies with 3 fresh training seeds each (plus committed
022 seed-0): ΔA is positive for every seed on both back-
023 bones (7B: $+0.0506 \pm 0.0046$; 2B: $+0.0316 \pm 0.0009$),
024 ΔG is positive for every seed (7B: $+0.0440 \pm 0.0054$;
025 2B: $+0.0316 \pm 0.0026$), while fresh-seed pair consistency
026 stays within 0.05 of the legacy checkpoint in both fam-
027 ilies and all fresh 2B answer-update rates exceed the
028 2B zero-shot rate. A post-confirmatory extension on
029 the contemporary Qwen3-VL-8B backbone reproduces
030 the primary $\Delta A/\Delta G$ pattern ($+0.032/+0.036$) and the
031 transformed-accuracy gain ($+0.045$). Ordinary spatial
032 fine-tuning is therefore not a scalar capability update:
033 benchmark accuracy, evidence dependence, and seman-
034 tic consistency must be measured—and reported—as
035 one vector whose components replicate across seeds
036 and families. The frozen protocol (conditions, met-

ric definitions, and independently recomputed audit
scripts) accompanies this paper so that future spatial-
adaptation results can be decomposed and compared
on equal terms.

1. Introduction

Spatial reasoning is central to multimodal systems that
operate in physical environments: a useful VLM must
determine whether one object is to the left of another,
above it, in front of it, or facing it, and it must do
so using the image—not a language prior. A grow-
ing line of work shows that benchmark accuracy can
diverge from genuine visual grounding: models can
answer correctly while depending weakly on the im-
age [2, 6, 7], and consistency across conditions can co-
exist with error [1]. Prior work has also introduced
grounding-aware training objectives that explicitly re-
ward visual dependence [5].

What is missing is an evaluation practice, not merely
another observation: spatial fine-tuning studies re-
port the scalar benchmark delta and leave evidence
dependence and semantic consistency unmeasured, so
two adaptations with identical accuracy gains can be
silently different in what they changed. We address
this gap by treating ordinary spatial task fine-tuning—
the standard LoRA recipe with no grounding-aware
objective—as a controlled intervention and measuring
it with a preregistered protocol that reports benchmark
accuracy, correct-image dependence, and semantic pair
consistency as one jointly measured vector of capability
changes, evaluated across training seeds and backbone
families.

We use Visual Spatial Reasoning (VSR) [3] as the
task. VSR contains natural-image statements about
object-relative relations (left/right, above/below, fac-
ing, containment, and others) paired with true/false
labels. Following the frozen protocol of the underlying

study (documented in the repository decision log), we evaluate every checkpoint under a corrected battery of six conditions: normal (the frozen prompt and image), shuffle (frozen derangement of the image), relcomp (strict complement pairs: language changes, expected label flips), facingcomp (facing-antonym flip-law diagnostic), and the Tier-C pair `hflip_flip` / `hflip_invariant` (global horizontal reflection; language fixed; left/right truth must flip, vertical/depth truth must stay).

We train three fresh General-LoRA seeds per backbone (7B Qwen2-VL-7B and 2B SmolVLM2-2B) using the identical recipe as the committed seed-0 run, then measure, per seed:

- ΔA (benchmark gain): normal accuracy after tuning minus zero-shot normal accuracy;
- ΔG (correct-image dependence): the change in the normal-minus-shuffle accuracy gap relative to zero-shot;
- ΔC (semantic pair consistency): relcomp/facingcomp flip-law compliance and its seed stability;
- Tier-C transformation behavior: three separate quantities under global reflection—transformed accuracy ($A_{\text{transform}}$), pair consistency (C_{pair}), and joint correctness (both_correct)—following the frozen metric definitions of the underlying study.

Every headline number in this paper was recomputed independently from raw prediction artifacts by a standalone audit script (scripts/audit_paper2_numbers_independent.py) that imports no analyzer module; the audit passed with zero discrepancies and zero failed claims. We report means and standard deviations across the three fresh seeds; seeds are independent training draws, not ordered stages.

Our primary finding is that the adaptation decomposes cleanly and reproducibly. ΔA is positive for all six fresh seeds across both families; ΔG is positive for all six; semantic pair consistency stays within 0.05 of the legacy General checkpoint for every fresh seed; and all fresh 2B Tier-C answer-update rates exceed the 2B zero-shot rate. A post-confirmatory extension on the contemporary Qwen3-VL-8B backbone reproduces the primary $\Delta A/\Delta G$ pattern and the transformed-accuracy gain, and is explicitly labeled exploratory rather than preregistered.

We make three contributions:

- A preregistered capability-vector protocol for auditing spatial fine-tuning: six frozen evaluation conditions and three jointly reported metrics (benchmark gain, correct-image dependence, semantic pair consistency) plus Tier-C transformation behavior, released with frozen manifests, metric definitions, and

an independent audit script, so that any spatial-adaptation study can report decomposed rather than scalar results.

- A controlled multi-seed, cross-backbone application of the protocol to ordinary LoRA fine-tuning on VSR: every vector component replicates across three fresh training seeds on two backbone families, closing the single-checkpoint-anecdote criticism.
- A scoped contemporary-backbone extension (Qwen3-VL-8B) that reproduces the primary adaptation pattern, honestly labeled as post-confirmatory external validation.

2. Related Work

Accuracy versus visual grounding. A central and rapidly growing observation is that benchmark accuracy can overstate genuine visual dependence. Beyond Accuracy (Zafar et al., 2026) [6] evaluates medical VLMs with real, blank, and shuffled images and defines the Visual Reliance Score (VRS) as $A(\text{real}) - A(\text{shuffle})$; they find that RLVR can improve accuracy while degrading visual grounding. Our correct-image dependence (the normal-minus-shuffle gap, G) is the same construction; we do not claim it as novel, and we report the zero-shot-relative change ΔG as the adaptation quantity. Seeing without Looking (Lan et al., 2026) [2] applies global and localized visual interventions to show that predictions can remain robust after evidence removal; it audits pretrained models rather than a fine-tuning transition. VisualFLIP (Zhu et al., 2026) [7] introduces a paired benchmark where the question stays fixed and a minimal task-critical edit deterministically flips the gold answer, with pair accuracy and Collapse Rate as metrics; its dataset and harness are not yet publicly released, and we do not claim to reproduce its protocol. Our Tier-C horizontal reflection is a global image intervention, not a minimal local edit; we report our pair metrics as collapse-style paired answer-update metrics following VisualFLIP, with the intervention difference stated explicitly.

Consistency versus grounding. Consistent Yet Wrong (Bhat and Yamasaki, 2026) [1] introduces ViewDiag and shows that multi-view consistency can coexist with substantial error, indicating prior-driven collapse rather than evidence-sensitive reasoning. Their dissociation between consistency and grounding is the static analogue of our ΔC -versus- ΔG decomposition, but they do not study a fine-tuning transition. VisualFLIP’s collapse rate is a similar paired-consistency notion for minimal-edit pairs.

175 Grounding-aware training objectives. CORAL (Zafar
176 et al., 2026) [5] fine-tunes Qwen2.5-VL-7B with a Con-
177 trastive Grounding Objective that penalizes answer in-
178 variance under hard-negative image swaps, improving
179 accuracy and reducing visual hallucination in medical
180 VQA. Med-OPD (Qian et al., 2026) [4] adds evidence-
181 aware on-policy distillation for medical VLMs. These
182 works design grounding-aware objectives; we study
183 what ordinary fine-tuning already changes, without
184 modifying the objective.

185 Positioning. No prior work, to our knowledge, treats
186 ordinary spatial task fine-tuning as a controlled inter-
187 vention and measures ΔA , ΔG , and ΔC (plus Tier-
188 C transformation behavior) jointly, per training seed,
189 across two backbone families, with a contemporary-
190 backbone extension. Our novelty claim is precisely
191 this joint, seed-robust decomposition—not the normal-
192 minus-shuffle gap, not “accuracy differs from ground-
193 ing,” and not a new training objective.

194 3. Protocol

195 Task and data. We use Visual Spatial Reasoning
196 (VSR) [3], test split ($n = 2195$ rows). Statements are
197 true/false sentences about object-relative relations; la-
198 bels are frozen. The frozen prompt template, gener-
199 ation settings (greedy, `max_new_tokens = 5`), im-
200 age preprocessing (uniform 392 px long-side cap), and
201 cache keys (md5 of the image URL) are identical to
202 the underlying committed protocol (documented in the
203 repository decision log and protocol freeze documents).

204 Evaluation battery (corrected, frozen). Following the
205 battery-drift correction in the decision log, the seed
206 campaign is evaluated with the already-committed
207 legacy grounding drivers, not the drifted heavy battery.
208 The corrected battery has six conditions with canonical
209 frozen subset sizes:

210 condition
normal / shuffle / blank / text_only
211 relcomp (strict complements)
facingcomp (facing-antonym)
hflip_flip (global reflection, flip-expected)
hflip_invariant (global reflection, expected-invariant)

212 The shuffle condition uses the frozen cycle-derangement
213 mapping (seed 20260810); identity and eligibility come
214 from frozen eligible-ids manifests. Attention is eager
215 for Qwen2-VL-7B and sdpa for SmolVLM2-2B, match-
216 ing every committed legacy run.

family	checkpoint	ΔA	G	ΔG
Qwen2-VL-7B	seed-0	+0.0542	0.3522	+0.0456
Qwen2-VL-7B	seedA	+0.0547	0.3544	+0.0478
Qwen2-VL-7B	seedB	+0.0456	0.3444	+0.0378
Qwen2-VL-7B	seedC	+0.0515	0.3531	+0.0465
SmolVLM2-2B	seed-0	+0.0287	0.2975	+0.0305
SmolVLM2-2B	seedA	+0.0305	0.3007	+0.0337
SmolVLM2-2B	seedB	+0.0323	0.2957	+0.0287
SmolVLM2-2B	seedC	+0.0319	0.2993	+0.0323
Qwen3-VL-8B	tuned	+0.0323	0.3827	+0.0355

Table 1. Headline adaptation quantities. $G = A_{\text{normal}} - A_{\text{shuffle}}$; ΔG is relative to zero-shot. All values from the independent audit.

217 Training recipe (verbatim seed-0 recipe). Each fresh
218 seed is a General-LoRA run on the general training
219 manifest (2000 rows; 95/5 split frozen at seed 42) with
220 the seed-0 recipe: epochs 2, lr 10^{-4} , LoRA $r=8$, $\alpha=16$,
221 dropout 0.05, bf16, gradient checkpointing. The only
222 intentional difference from seed-0 is the explicit per-
223 run seed (101/202/303) applied to torch/cuda/numpy
224 at driver start. Three seeds per backbone were trained
225 on A6000 GPUs.

226 Fresh-seed analysis. Seeds are independent training
227 draws, not ordered stages; we report per-seed values,
228 means, and standard deviations, and never “monotonic
229 across seeds.” All comparisons are within-backbone;
230 preprocessing is identical within a backbone.

231 Independent numerical audit. Every
232 headline quantity in this paper was
233 recomputed by a standalone script
234 (scripts/audit_paper2_numbers_independent.py)
235 that imports no analyzer module and reads only raw
236 prediction CSVs/JSONL, frozen protocol manifests,
237 and committed analysis JSONs as comparison targets.
238 The audit hard-fails on any deterministic-proportion
239 discrepancy above 10^{-6} , any row-count/ID mismatch,
240 and any terminology mismatch. It passed with zero
241 n issues, and the committed seed-0 tier-a values were
242 produced byte-identically (7B normal 0.82414579;
243 660B normal 0.76492027; absolute difference 0.0).

244 4. Results

245 All values below are audit-sourced (Step-1 independent
246 audit, PASS). Seeds are reported per-draw; means/SDs
247 are over the three fresh seeds.

248 Benchmark gain (ΔA) and correct-image dependence
249 (ΔG). Table 1 reports the per-seed quantities. ΔA

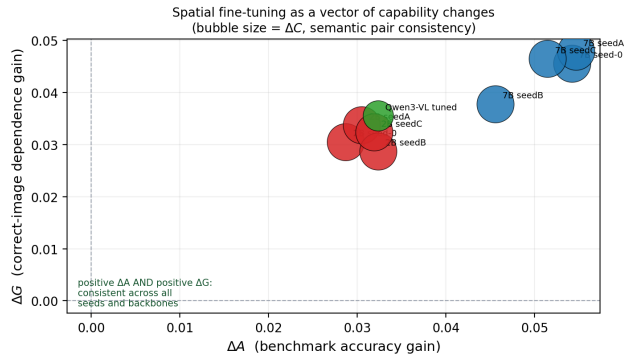


Figure 1. Conceptual decomposition: ΔA (x) versus ΔG (y) for all nine checkpoints (7B seed-0 + three seeds, 2B seed-0 + three seeds, Qwen3-VL-8B tuned); bubble size encodes ΔC (semantic pair consistency change). All points lie in the positive-positive quadrant.

family	checkpoint	C_{pair} (relcomp)
Qwen2-VL-7B	seed-0	0.6772
Qwen2-VL-7B	seedA	0.6637
Qwen2-VL-7B	seedB	0.6547
Qwen2-VL-7B	seedC	0.6652
SmolVLM2-2B	seed-0	0.5015
SmolVLM2-2B	seedA	0.4985
SmolVLM2-2B	seedB	0.5045
SmolVLM2-2B	seedC	0.5105

Table 2. Relcomp pair consistency ($n = 666$). Fresh seeds stay within 0.05 of the legacy General checkpoint in both families.

is positive for every seed on both backbones (7B fresh-seed mean $+0.0506 \pm 0.0046$; 2B fresh-seed mean $+0.0316 \pm 0.0009$). The normal-minus-shuffle gap G widens under tuning in every family, and the zero-shot-relative change ΔG is positive for every seed (7B fresh-seed mean $+0.0440 \pm 0.0054$; 2B fresh-seed mean $+0.0316 \pm 0.0026$). Shuffle accuracy itself stays flat (e.g., 2B zero-shot 0.4692 vs. tuned 0.4674–0.4729), so the gain is in using the correct image, not in memorizing image-independent answers.

Semantic pair consistency (ΔC , relcomp). Table 2 shows relcomp C_{pair} . Fresh seeds cluster tightly around seed-0 in both families (7B: 0.6547–0.6652 vs. 0.6772; 2B: 0.4985–0.5105 vs. 0.5015); the audit confirms every fresh seed is within 0.05 of the legacy General checkpoint. The facing-antonym diagnostic (facingcomp, $n = 103$) shows the same pattern of tight seed clustering (2B seed-0 0.3469; fresh seeds 0.3429–0.3714).

family	checkpoint	$A_{\text{transform}}$	C_{pair}	both_correct
Qwen2-VL-7B	zero-shot	0.6367	0.6163	0.5388
Qwen2-VL-7B	seed-0	0.6571	0.6857	0.5959
Qwen2-VL-7B	seedA	0.6490	0.6490	0.5796
Qwen2-VL-7B	seedB	0.6571	0.6898	0.6041
Qwen2-VL-7B	seedC	0.6449	0.6653	0.5837
SmolVLM2-2B	zero-shot	0.4980	0.3184	0.2531
SmolVLM2-2B	seed-0	0.5224	0.3469	0.2980
SmolVLM2-2B	seedA	0.5469	0.3429	0.3061
SmolVLM2-2B	seedB	0.5469	0.3633	0.3143
SmolVLM2-2B	seedC	0.5469	0.3714	0.3224

Table 3. Tier-C hflip_flip ($n = 245$). $A_{\text{transform}}$ is transformed-answer accuracy; C_{pair} is the answer-update rate ($\text{mirrored} \neq \text{normal}$); both_correct is joint correctness. Three separate quantities, never collapsed.

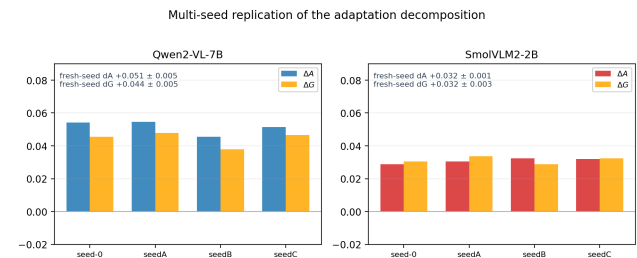


Figure 2. Multi-seed replication: per-seed ΔA and ΔG for both confirmatory families, with fresh-seed mean $\pm$ SD annotations.

Transformation behavior under global reflection (Tier C). Table 3 reports the three Tier-C quantities separately. Under global horizontal reflection (language fixed, left/right truth flips):

- Transformed accuracy ($A_{\text{transform}}$): 2B rises from 0.4980 (zero-shot) to 0.5224 (seed-0) and 0.5469 for each fresh seed; 7B 0.6367 $\rightarrow$ 0.6571 with fresh seeds 0.6490–0.6571.
- Answer-update rate (C_{pair}): 2B 0.3184 (zero-shot) $\rightarrow$ 0.3469 (seed-0), with all three fresh seeds higher than zero-shot (0.3429/0.3633/0.3714); 7B 0.6163 $\rightarrow$ 0.6857, fresh seeds 0.6490/0.6898/0.6653.
- Joint correctness (both_correct): 2B 0.2531 $\rightarrow$ 0.2980 (seed-0) $\rightarrow$ 0.3061/0.3143/0.3224.

The audit confirms all fresh 2B C_{pair} values exceed the 2B zero-shot value, and all fresh-seed C_{pair} values (both families) are within 0.05 of the legacy General checkpoint.

Contemporary-backbone extension (Qwen3-VL-8B, post-confirmatory). Table 1 includes the Qwen3-VL-8B row: $\Delta A +0.0323$, $\Delta G +0.0355$, and hflip_flip transformed accuracy $+0.0449$ (0.6571 $\rightarrow$ 0.7020).

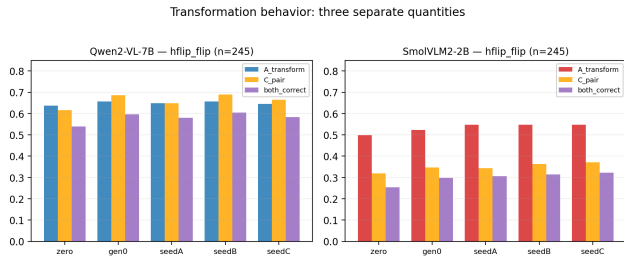


Figure 3. Transformation behavior: $A_{\text{transform}}$, C_{pair} , and both_correct per checkpoint, hflip_flip ($n = 245$), both families.

This extension is explicitly exploratory: it computed only ΔA , ΔG , and $A_{\text{transform}}$; C_{pair} was not computed for it, and no response-law-compliance claim is made. It is labeled post-confirmatory external validation, not preregistered confirmatory evidence.

Claim-level summary (all audited).

- $\Delta A > 0$ for all fresh seeds, both confirmatory backbones: PASS.
- $\Delta G > 0$ for all fresh seeds, both confirmatory backbones: PASS.
- Fresh-seed C_{pair} within 0.05 of legacy General (7B and 2B): PASS.
- All fresh 2B $\text{hflip_flip } C_{\text{pair}} > 2\text{B zero-shot}$: PASS.
- Qwen3-VL-8B supports only the stated $\Delta A/\Delta G/A_{\text{transform}}$ quantities: PASS.

5. Discussion and Limitations

What the decomposition shows. The headline result is that ordinary spatial fine-tuning is not a scalar capability update. Benchmark accuracy, correct-image dependence, and semantic pair consistency move together in direction (all positive for ΔA and ΔG ; ΔC stable) but they are distinct quantities: shuffle accuracy stays flat while the gap widens; answer-update rates improve while transformed accuracy moves differently; and the three Tier-C quantities never collapse into one number. A model can get the mirrored item right without updating its answer relative to its own original prediction, and vice versa; reporting them separately is essential.

Seed robustness. Every component replicates across three fresh training draws per family. The fresh-seed standard deviations are small (7B ΔA SD 0.0046; 2B ΔA SD 0.0009), which closes the single-checkpoint-anecdote criticism that a training effect could be a lucky draw.

Relationship to prior work. We do not claim the normal-minus-shuffle gap as novel: VRS in Beyond Accuracy [6] is the same construction. We do not claim to reproduce VisualFLIP [7]: our reflection is a global image intervention, and we state the difference. Our novelty is the joint, seed-robust, cross-backbone decomposition of ordinary fine-tuning into ΔA , ΔG , ΔC , and Tier-C behavior.

Limitations. (1) Two backbone families (7B and 2B) plus one contemporary extension; the extension computed only $\Delta A/\Delta G/A_{\text{transform}}$ and is exploratory. (2) The Tier-C intervention is a global horizontal reflection; VisualFLIP-style minimal local edits were not run because the official dataset is not yet public. (3) All comparisons are within-backbone with identical preprocessing; we do not claim resolution or architecture independence. (4) The shuffle condition tests dependence on the correct image, which we call correct-image dependence, not “causal grounding”; stronger causal language is reserved for genuine counterfactual evidence. (5) A text-only control adapter was considered but not run; it remains future work. (6) Seeds are independent draws; we do not claim a “monotonic” seed trend anywhere.

6. Conclusion

Ordinary spatial LoRA fine-tuning on VSR produces a reproducible vector of capability changes, not a single benchmark gain. Across three fresh training seeds and two backbone families, ΔA (benchmark accuracy) and ΔG (correct-image dependence) are positive for every seed, semantic pair consistency ΔC stays within 0.05 of the legacy General checkpoint, and Tier-C transformation behavior improves answer-update rates while remaining a set of three separately reported quantities. A contemporary Qwen3-VL-8B extension reproduces the primary $\Delta A/\Delta G$ pattern and the transformed-accuracy gain. Every number was independently recomputed from raw artifacts and the frozen protocol; the full claim hierarchy, numerical audit, literature audit, figures, and tables are archived in the repository. We recommend that spatial fine-tuning be evaluated and reported as this joint vector—accuracy, correct-image dependence, semantic consistency, and transformation behavior—rather than collapsed into a single accuracy number.

References

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